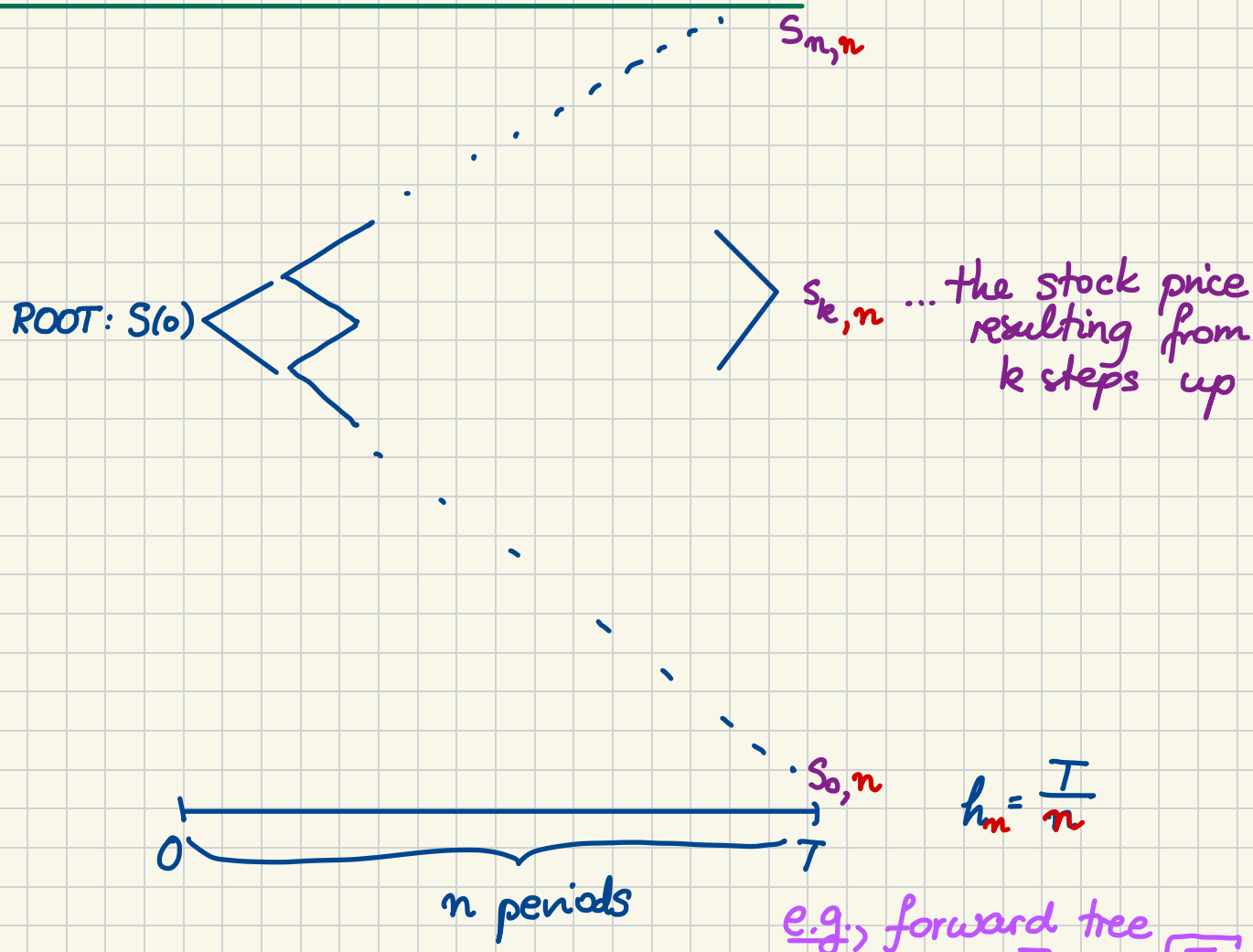
The Pre-Limit n-Period Binomial Tree.



$h_n = \frac{T}{n}$

u_n ... up factor

d_n ... down factor

e.g., forward tree

$u_n = \exp\left(r \cdot \frac{T}{n} + \sigma\sqrt{\frac{T}{n}}\right)$

$d_n = \exp\left(r \cdot \frac{T}{n} - \sigma\sqrt{\frac{T}{n}}\right)$

$$S_{k,n} = S(0) \cdot u_n^k \cdot d_n^{n-k} = S(0)\left(\frac{u_n}{d_n}\right)^{k} \cdot d_n^n$$

k .. corresponds to a realization of the binomial r.v. w/ n trials

and success probability

$$p_n^* = \frac{e^{r\left(\frac{T}{n}\right)} - d_n}{u_n - d_n}$$

e.g., in the forward tree

$$\boxed{p_n^* = \frac{1}{1+e^{\sigma\sqrt{h}}}} \xrightarrow[n\to\infty]{} \frac{1}{2}$$

Say, X_n .... # of steps up in n periods

$X_n \sim \text{Binomial}(\#\text{ of trials} = n,\ \text{success probab} = p_n^*)$

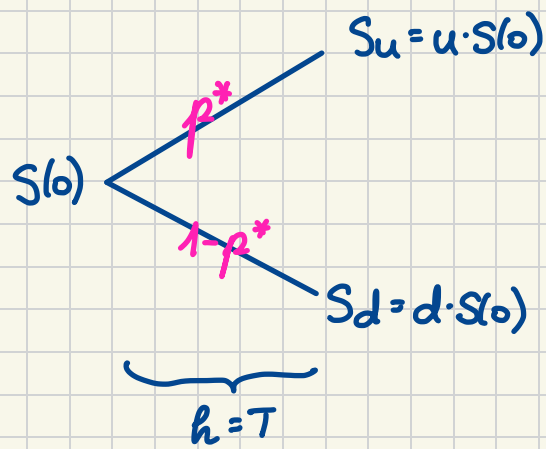
Q: Can we simply use the normal approximation to the binomial?

$Y_n \sim \text{Binomial}(n = \#\text{ of trials},\ p = \text{success probability})$

$$\boxed{\text{☹ } p_n^* \text{ depends on } n}$$

Subjective Probability.

Recall: When pricing we use the risk-neutral probability measure $\mathbb{P}^*$ associated w/ $\boxed{p^* = \dfrac{e^{rh} - d}{u - d}}$



Q: If we invest in one share of this non-dividend-paying stock @ time 0, what is the expected wealth @ time T under $\mathbb{P}^*$?

$\longrightarrow$: $\mathbb{E}^*[S(T)] = S(0)e^{rT}$ $\square$

In Contrast:

There can be a subjective probability measure $\mathbb{P}$.
We can think about the quality of our investment under that probability measure

$$\mathbb{E}[S(T)] = S(0)e^{\alpha T}.$$

We call this α the mean rate of return.
In a binomial tree, we can express the "true probability" of a step up as

$$\boxed{p = \frac{e^{\alpha h} - d}{u - d}}$$

The mean rate of return on the stock under the risk-neutral probability measure $\mathbb{P}^*$ is $\underline{r}$.